FitCheck — Entity-Relationship (ER) Conceptual Diagram

DBMS Experiential Learning Level 3 | Hybrid Relational-Vector Clothing Fit Recommendation Engine

MySQL 8.0 + Redis

Legend: Strong Entity | Dependent / Subordinate Entity | Associative / Transactional Entity | **[PK] Primary Key** | **[FK] Foreign Key** | 1 : 1 (One-to-One) | 1 : N (One-to-Many) | === Total Participation

USER (Strong Entity)

- **user_id : INT [PK]**
- name : VARCHAR(100)
- email : VARCHAR(255) [UQ]
- password_hash : VARCHAR(255)
- role : VARCHAR(10) [CHK]
- created_at : TIMESTAMP

Role: USER / ADMIN

1 : 0..1

BODY_PROFILE (Dependent)

- **profile_id : INT [PK]**
- **user_id : INT [FK, UQ] (CASCADE)**
- height : DECIMAL(5,2) [cm]
- waist : DECIMAL(5,2) [cm]
- hip : DECIMAL(5,2) [cm]
- thigh : DECIMAL(5,2) [cm]
- inseam : DECIMAL(5,2) [cm]
- preferred_fit : VARCHAR(20)
- updated_at : TIMESTAMP

1 : N

1 : N

PURCHASE (Associative)

- **purchase_id : INT [PK]**
- **user_id : INT [FK]**
- **garment_size_id : INT [FK]**
- purchase_date : DATE
- outcome : VARCHAR(10) [CHK]
- fit_satisfaction : VARCHAR(10) [CHK]
- fit_preference : VARCHAR(20) [CHK]

Outcome: KEPT / RETURNED / EXCHANGED

purchased size (1:N)

FIT_REVIEW (Associative)

- **review_id : INT [PK]**
- **user_id : INT [FK]**
- **garment_size_id : INT [FK]**
- overall_fit : VARCHAR(20) [CHK]
- waist_fit, hip_fit, thigh_fit : VARCHAR(10)
- length_fit : VARCHAR(10) [CHK]
- review_text : TEXT [CHK: non-empty]
- created_at : TIMESTAMP

⇄ Synchronized to Redis Vector Store (384d)

reviewed size

GARMENT_SIZE (Dependent)

- **garment_size_id : INT [PK]**
- **garment_id : INT [FK]**
- size_label : VARCHAR(20)

UQ(garment_id, size_label)

1 : N

1 : N

GARMENT_MEASUREMENT

- **measurement_id : INT [PK]**
- **garment_size_id : INT [FK] (CASCADE)**
- measurement_type : VARCHAR(20) [CHK]
- measurement_value : DECIMAL(6,2)
- unit : VARCHAR(5) [cm / in]

Types: waist, hip, thigh, inseam, rise

BRAND (Strong Entity)

- **brand_id : INT [PK]**
- brand_name : VARCHAR(100) [UQ]
- sizing_system : VARCHAR(20)

ALPHA / NUMERIC / WAIST_BASED

1 : N

GARMENT (Strong Entity)

- **garment_id : INT [PK]**
- **brand_id : INT [FK]**
- garment_name : VARCHAR(150)
- garment_category : VARCHAR(30)
- fit_type : VARCHAR(30) [CHK]
- description : TEXT

Category: Jeans / Trousers

Formal Entity Classification & Participation Constraints Rationale:

1. **Strong Entities (USER, BRAND, GARMENT):** Possess independent existence and autonomous surrogate primary keys (user_id, brand_id, garment_id).
2. **Dependent Subordinate Entities (BODY_PROFILE, GARMENT_SIZE, GARMENT_MEASUREMENT):** Subordinate lifecycle; GARMENT_SIZE discriminates size_label under garment_id (UQ); GARMENT_MEASUREMENT relies on garment_size_id with CASCADE deletion
3. **Associative Entities (PURCHASE, FIT_REVIEW):** Model M:N interactions between USER and GARMENT_SIZE with historical outcomes, fit ratings, and natural-language text.
4. **Strict Cross-User Privacy Boundary:** Recommendation calibration queries only the authenticated user's private BODY_PROFILE and PURCHASE records. Community FIT_REVIEW text is analyzed solely for target garment fit consensus without accessing reviewer body profiles.